MEHMET KEREM TURKCAN

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SUMMARY

Associate Research Scientist at Columbia University working on physical AI: models that perceive, simulate, and act in the physical world. I work on sim-to-real transfer for urban perception and robots, from the Constellation benchmark and Boundless synthetic data to Valdrada, a real-time digital twin of New York City, and humanoid navigation trained in Isaac Lab; on vision-language and open-vocabulary 3D perception at the edge; on vision-language-action and world models for medical robotics; and on LLM post-training and agents. My record includes systems deployed across major urban testbeds, publications at CVPR, ICML, ACM UIST, ACM/IEEE SEC, IEEE INFOCOM, IJCV, and eLife, and open-source projects with 2,200+ GitHub stars.

SELECTED CONTRIBUTIONS

- ◇ **Physical AI and sim-to-real:** Built Valdrada, a real-time digital twin of New York City with pixel-exact ground truth (258 GitHub stars in six weeks), and G1 Patrol, humanoid navigation through crowded sidewalks trained in Isaac Lab; Boundless synthetic data outperforms CARLA renders for training detectors on real cameras.
- ◇ **Real-world AI systems:** Lead multimodal perception and urban analytics efforts intended to scale across 900+ New York City intersections and the COSMOS PAWR, DataCity, and FAU MobIntel testbeds.
- ◇ **Frontier model deployment:** Created DART, a real-time open-vocabulary detector with TensorRT deployment, public weights, and 790+ GitHub stars; led UrbanOmniDetect, an oral paper at the CVPR 2026 DriveX Workshop.
- ◇ **Sponsored research:** Contributed proposal development, technical reporting, sponsor reviews, and engineering delivery across approximately $30M in institutional research supported by NSF, DARPA, AFOSR, and Con Edison, with additional project support from NVIDIA and EmpireAI.
- ◇ **Teaching and mentorship:** Taught Columbia graduate deep learning courses, mentored Master's, undergraduate, and high school researchers, six of whom went on to Ph.D. programs, and served as the main engineering instructor for CS3's Research Experience for Teachers in 2024–2026.

EXPERIENCE

◇ **Columbia University**, New York, NY — Jul 2024 – Present
Associate Research Scientist, Civil Engineering & Engineering Mechanics

- Lead large-scale applied AI projects in physical AI and robot learning, real-time video analytics, open-vocabulary and 3D perception, edge/VLM systems, urban digital twins, and medical robotics foundation models; manage shared high-performance computing and cross-team engineering.
- Created DART, converting promptable frontier vision models into a real-time multiclass detector through shared backbone inference, batched decoding, adapter distillation, and TensorRT FP16 optimization.
- Led UrbanOmniDetect, a calibration-free, view-agnostic monocular 3D detector for ego-vehicle, infrastructure, and aerial cameras, with public models and the UrbanOmniView dataset.
- Lead work toward a civic robot system for CS3: Unitree G1 humanoid and B2-W quadruped navigation in dense pedestrian environments, trained in Isaac Lab with PPO teacher-student distillation, domain randomization, and a ROS 2 deployment stack.
- Lead surgical world model research for healthcare robotics and contributed to Open-H-Embodiment; developed public model assets for video generation, phase recognition, metric depth, and skills assessment.
- Mentored the student first authors of papers at ICML 2025, IEEE INFOCOM 2026, and USENIX VehicleSec 2026; co-authored the ACM/IEEE SEC 2025 Best Paper on privacy-preserving urban safety analytics.
- Designed and led the engineering curriculum for three annual CS3 Research Experience for Teachers cohorts, translating current AI research into reproducible K–12 lesson plans and classroom projects.

◇ **Columbia University**, New York, NY — Nov 2022 – Jul 2024
Postdoctoral Research Scientist

- Built real-time digital twins of New York City intersections integrating object detection, multi-object tracking, trajectory forecasting, simulation, and safety warning generation.
- Developed synthetic data and benchmarking systems for dense urban perception, including Boundless in Unreal Engine 5 and the Constellation high-altitude object detection dataset.
- Built healthcare AI pipelines with Northwell Health collaborators for surgical phase recognition and automated endoscopy skill scoring, resulting in peer-reviewed journal publications and clinical presentations.
- Mentored 11 Master's researchers and three high school researchers across computer vision, generative modeling, and urban analytics projects.

◇ **Columbia University**, New York, NY Jan 2023 – Dec 2023
Lecturer in Electrical Engineering
- Designed and taught ECBM E4040 Neural Networks & Deep Learning and EECS E6691 Advanced Deep Learning, including lectures, coding assignments, exams, and conference-style final projects.
- Co-taught VOAI 0003E Neural Networks and Deep Learning for the Columbia Video Network AI Executive Certificate in Fall 2022 and Summer 2023.

◇ **Columbia University**, New York, NY Jan 2017 – Oct 2022
Doctoral Researcher, Bionet Group
- Co-created FlyBrainLab, an open computing platform for natural language exploration, graph querying, simulation, and visualization of connectome-scale neuroscience data.
- Developed an early research agent architecture combining programmable ontologies, entity retrieval linked to the literature, dense passage retrieval, biomedical BERT question answering, graph database tools, and executable scientific workflows.
- Built scalable TypeScript and Python interfaces, OrientDB/NeuroArch query services, neural simulations on GPUs, and interactive 3D tools used in open neuroscience research and education.

◇ **Columbia University**, New York, NY 2015 – 2016
Graduate Researcher
- Developed deep learning systems for thermal camera infrastructure inspection, solar microgrid control, and intelligent building forecasting.

EDUCATION

◇ **Columbia University**, New York, NY 2022
- Ph.D. in Electrical Engineering; Systems Biology and Neuroengineering, GPA 4.10.

◇ **Columbia University**, New York, NY 2016
- M.S. in Computer Science; Machine Learning and Thesis Track, GPA 3.97.

◇ **Istanbul Technical University**, Istanbul, Turkey 2015
- B.S. in Electronics and Communication Engineering, GPA 3.75.

AWARDS AND RECOGNITION

◇ NVIDIA research grant, *Generative Edge AI for Dynamic Smart Urban Metropolises*, with A. Smyth 2026
◇ **Best Paper Award**, ACM/IEEE Symposium on Edge Computing 2025
◇ NVIDIA research grant, *Edge AI for Equitable and Safe Intersections in Urban Metropolises*, with Z. Kostic 2025
◇ EmpireAI award, *AI Computing for Community, Health, Safety and Climate* 2025
◇ INRIX x MetroLab Challenge finalist 2025
◇ Columbia University Electrical Engineering Distinguished Teaching Award 2024
◇ Smart Cities North America Award winner in Transportation, IDC Government Insights 2023
◇ National Postdoc Appreciation Week Excellence Award, Columbia University 2023
◇ Herbert French Fellowship, Columbia University; Helmsley Fellowship, CSHL 2017–2019

SPONSORED RESEARCH CONTRIBUTIONS

Institutional award totals; dates indicate my period of contribution.

◇ **NSF Engineering Research Center for Smart Streetscapes (CS3)** (EEC-2133516) 2022 – Present
Research staff; research coordination, annual reporting, and sponsor reviews $26M center award
- Coordinate applied AI research and cross-team demonstrations across a five-institution center; present DART, UrbanOmniDetect, and related systems at annual NSF site visits.

◇ **NSF CPS: Hybrid Twins for Urban Transportation** (CNS-2038984) 2021 – Present
Technical, reporting, and sponsor review lead $1.3M Columbia award
- Led engineering delivery, prepared annual technical reports, and presented project reviews to NSF for traffic sensing, prediction, and digital twin systems spanning individual intersections and citywide operations.

◇ **DARPA Robust Learning in Brain Circuits of Synthetic Miniature Insects** (HR0011-19-9-0035) 2019 – 2020
Proposal and technical report contributor $620K Columbia award
- Proposal and technical report contributor; contributed to proposal development, authored project reports, and conducted grant-supported doctoral research on robust learning and executable brain circuit models.

◇ **AFOSR Foundations of Neuroinformation Processing** (FA9550-16-1-0410) 2017 – 2021
Grant-supported doctoral researcher $1.11M Columbia award
- Developed computational systems and models for phase- and spike-based neural information processing, including work underlying the FlyBrainLab platform.

◇ **NSF Collaborative Research: The Digital Fly Brain** (DBI-1544383) 2017 – 2019
Grant-supported doctoral researcher $795K Columbia award

- Engineered connectome data, graph query, simulation, and visualization capabilities for the open-source Fruit Fly Brain Observatory and FlyBrainLab systems.

◇ **Con Edison Underground Infrastructure and Public Safety Analytics** (CU09-1331) 2015 – 2016
Graduate Researcher; Amendments 21–23 Con Edison

- Developed theoretical models, implementations, and data processing pipelines for a Visual Data Capture pilot using thermal imagery, high-resolution photographs, and video of underground electrical structures; examined automated hotspot detection, inspection targeting, and estimation of avoided serious events.
- Extended the system with infrared camera data assimilation, equipment and cable density models, temperature differential extraction along cable lengths, and visual features associated with later failures; delivered project outputs to Con Edison.
- Contributed cost-benefit and portfolio optimization analysis for secondary public safety programs.

SELECTED SOFTWARE AND DATA

◇ **Valdrada:** Real-time digital twin of New York City from public records, with simulated traffic and pedestrians, pixel-exact ground truth, and a Python API modeled on CARLA. github.com/mkturkcan/valdrada
◇ **G1 Patrol:** Humanoid and quadruped navigation in dense pedestrian environments, trained in Isaac Lab with teacher-student distillation. github.com/mkturkcan/g1_patrol
◇ **DART:** Real-time open-vocabulary detection; PyTorch, TensorRT, distillation, public weights, and deployment benchmarks. github.com/mkturkcan/DART
◇ **UrbanOmniDetect:** Calibration-free monocular 3D detection with public models and the UrbanOmniView multi-view dataset. github.com/mkturkcan/urbanomnidetect
◇ **Loom:** Analytical neural computer that runs compiled C programs inside a looped transformer, with ONNX/WebGPU demos, FPGA verification, and 140+ tests. github.com/mkturkcan/Loom
◇ **FlyBrainLab:** Natural language and ontology interfaces, OrientDB graph querying, GPU simulation, and 3D scientific visualization. github.com/FlyBrainLab/FlyBrainLab
◇ **Boundless:** Unreal Engine 5 synthetic data for infrastructure and aerial detection that outperforms CARLA renders on real cameras. github.com/mkturkcan/boundless
◇ **bikeped:** Real-time fisheye perception, safety alerts, simulator scenarios, and conformance testing for urban intersections. github.com/mkturkcan/bikeped

SELECTED PUBLICATIONS AND PREPRINTS

1. **Constellation Dataset: Benchmarking High-Altitude Object Detection for an Urban Intersection. Mehmet Kerem Turkcan**, C. Zang, S. Narasimhan, G. H. Je, B. Yu, M. Ghasemi, J. Ghaderi, G. Zussman, and Z. Kostic. *International Journal of Computer Vision, 134(10):429, 2026.*
2. **Calibration-Free View-Agnostic Monocular 3D Object Detection for Urban Scenes. Mehmet Kerem Turkcan**, D. Gumaste, and Z. Kostic. *CVPR Workshops (DriveX), oral presentation, 2026.*
3. **AI-Powered CPS-Enabled Vulnerable-User-Aware Urban Transportation Digital Twin: Methods and Applications.** Y. Fu, **Mehmet Kerem Turkcan**, M. Ghasemi, Z. Mo, C. Zang, A. Adhikari, Z. Kostic, G. Zussman, and X. Di. *IEEE Transactions on Intelligent Transportation Systems, 27(5):4949–4966, 2026.*
4. **Worst-Case Attacks in Reactive Edge-Cloud Systems: Expected Latency and SLA Violation.** J. Tavori, **Mehmet Kerem Turkcan**, Z. Kostic, J. Ghaderi, and G. Zussman. *IEEE INFOCOM, 2026.*
5. **Digital Eyes on the Road: Using Street Cameras to Verify Traffic Integrity and Detect Sybil Attacks at City Scale.** J. Tavori, **Mehmet Kerem Turkcan**, Z. Kostic, S. Stolfo, and G. Zussman. *USENIX VehicleSec, 2026.*
6. **Harnessing Floating Car Data, Traffic Camera Observations, and Network Flow Analysis for Traffic Volume Estimation.** A. Kosikova, **Mehmet Kerem Turkcan**, A. Darrat, and A. Smyth. *Discover Civil Engineering, 3(1):257, 2026.*
7. **Newell-Car-Following-Model-Informed Gaussian Process Regression Model for Ultra-Short-Term Arterial Vehicle Trajectory Prediction.** N. S. Ahmad, P. J. Jin, C. Zang, **Mehmet Kerem Turkcan**, and Z. Kostic. *Transportation Research Record, to appear, 2026.*
8. **Zero-Shot Image Tagging at Traffic Intersections via SAM-Guided Vision-Language Models.** W. Ho, **Mehmet Kerem Turkcan**, J. Ghaderi, G. Zussman, and Z. Kostic. *ICMLA, 2026.*
9. **Computer Vision Scoring of Endoscopically Traced Line Figures in an Inanimate Scope Tip Coordination Training Model.** N. Mitra, S. Narasimhan, **Mehmet Kerem Turkcan**, et al. *Surgical Endoscopy, 40(1):657–664, 2026.*
10. **Dynamic Minimax Regret Optimization for Robust LLM Post-Training.** C. Zang, H. Dong, **Mehmet Kerem Turkcan**, G. Zussman, Z. Kostic, and J. Ghaderi. *arXiv:2610.06329, 2026.*
11. **Surgical Kinematics from Monocular Video with Learned Articulated Motion Constraints. Mehmet Kerem Turkcan**, S. Samal, and Z. Kostic. *arXiv:2609.27227, 2026.*
12. **Open-H-Embodiment: A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics.** N. Nelson, J.-T. Chen, J. Haworth, et al., including **Mehmet Kerem Turkcan**. *arXiv:2604.21017, 2026.*

13. **A Vision-Based Analysis of Congestion Pricing in New York City. Mehmet Kerem Turkcan**, J. Tavori, J. Ghaderi, G. Zussman, Z. Kostic, and A. Smyth. *arXiv:2602.03015, 2026.*
14. **Real-Time Video Analytics for Urban Safety: Deployment over Edge and End Devices.** M. Ghasemi, Y. Fu, X. Ouyang, P. Wang, **Mehmet Kerem Turkcan**, et al. *ACM/IEEE SEC, Best Paper Award, 2025.*
15. **Adaptive Data Collection for Robust Learning Across Multiple Distributions.** C. Zang, **Mehmet Kerem Turkcan**, G. Zussman, Z. Kostic, and J. Ghaderi. *ICML, 2025.*
16. **Distributed VLMs: Efficient Vision-Language Processing through Cloud-Edge Collaboration.** Y. Li, D. Gumaste, **Mehmet Kerem Turkcan**, J. Ghaderi, G. Zussman, and Z. Kostic. *IEEE PerCom Workshops, 2025.*
17. **Towards Suturing World Models: Learning Predictive Models for Robotic Surgical Tasks. Mehmet Kerem Turkcan**, M. Ballo, F. Filicori, and Z. Kostic. *arXiv:2503.12531, 2025.*
18. **StreetNav: Leveraging Street Cameras to Support Precise Outdoor Navigation for Blind Pedestrians.** G. Jain, B. Hindi, Z. Zhang, et al., including **Mehmet Kerem Turkcan**. *ACM UIST, 2024.*
19. **Boundless: Generating Photorealistic Synthetic Data for Object Detection in Urban Streetscapes. Mehmet Kerem Turkcan**, Y. Li, C. Zang, J. Ghaderi, G. Zussman, and Z. Kostic. *arXiv:2409.03022, 2024.*
20. **A Programmable Ontology Encompassing the Functional Logic of the Drosophila Brain. Mehmet Kerem Turkcan**, A. A. Lazar, and Y. Zhou. *Frontiers in Neuroinformatics, 2022.*
21. **Accelerating with FlyBrainLab the Discovery of the Functional Logic of the Drosophila Brain. Mehmet Kerem Turkcan**, A. A. Lazar, T. Liu, and Y. Zhou. *eLife, 2021.*

TEACHING AND CURRICULUM LEADERSHIP

◇ **Main Engineering Instructor, CS3 Research Experience for Teachers** 2024 – 2026
Center for Smart Streetscapes New York, NY
- Designed and led a four-week curriculum covering neural networks, model evaluation, computer vision, object tracking, generative AI, annotation, YOLO training, and edge deployment; supported teachers in producing classroom-ready lesson plans.

◇ **Course and Teaching Support, Columbia University** 2017 – 2021
Computational neuroscience and neuroengineering courses New York, NY
- Supported Computing with Brain Circuits of Model Organisms, Fruit Fly Brain as a Neuroinformation Processor, and Computational Neuroscience.

SELECTED PRESENTATIONS

◇ **CS3 NSF Annual Site Visit** May 12–15, 2026
DART and UrbanOmniDetect research demonstrations Columbia University
- Presented multiple live applied AI systems during the center's annual NSF review.

◇ **From Sensors to Systems: Real-Time Traffic Analysis for Faster Decision-Making** Feb. 5, 2026
Panelist, CS3 Innovation Summit Columbia University
- Discussed real-time sensing, edge AI, and operational urban analytics for faster public-sector decision-making.

◇ **Port Authority Tomorrow Summit** Jan. 7, 2026
Workshop host World Trade Center, New York
- Led a technology workshop at the Port Authority of New York and New Jersey headquarters.

◇ **Evaluating Micromobility & Dangerous Riding Behaviors** Nov. 19, 2025
Eighth Annual Vision Zero Research on the Road New York, NY
- Presented camera-based methods for evaluating micromobility patterns and dangerous riding behavior to New York City transportation stakeholders.

PROFESSIONAL SERVICE

Organizing committee: Poster and Demo Co-Chair, IEEE SMARTCOMP 2027.
Peer reviewer: CVPR 2026; ICML 2026; ICMLA 2026; WACV 2025; ACM MobiCom 2024; IEEE/ACM ToN; IEEE TVLSI; Sensors, Electronics, and Smart Cities (MDPI).

TECHNICAL SKILLS

◇ **AI and deployment:** PyTorch, JAX, TensorFlow, Hugging Face, Ultralytics, OpenCV, TensorRT, ONNX, LiteRT, NCNN, WebGPU; NVIDIA Jetson, Qualcomm, and Raspberry Pi edge devices; detection, segmentation, tracking, 3D perception, VLM/VLA systems, LLM post-training (DPO, RL), RAG, agents, diffusion, and world models.
◇ **Simulation and robotics:** NVIDIA Isaac Lab and Isaac Sim, Newton (MuJoCo-Warp), Unreal Engine 5 (UE5), CARLA, OpenUSD, Blender, ROS 2; humanoid and quadruped robot learning for the Unitree G1 and B2-W (PPO, teacher-student distillation, domain randomization); sim-to-real transfer, city-scale digital twins, traffic and pedestrian simulation, synthetic data, and hardware-in-the-loop testing.
◇ **Languages, systems, and data:** Python, C/C++, CUDA, TypeScript/JavaScript, Node.js, SQL, MATLAB; OrientDB, RPC, distributed querying, HPC, Docker, GitHub Actions; large-scale data collection and annotation (CVAT), OpenStreetMap and NYC Open Data, JupyterLab, Three.js/WebGL, scientific visualization, and Adobe Creative Suite.